Literature-Implied Answer to Morillon's Asymptotically Isometric ℓ^1 Question

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Status

Literature-implied answer, ZFC variant of Question 3. This note records that a theorem of Pfitzner answers, in ZFC, the third question in Marianne Morillon's *A new proof of James' sup theorem*, arXiv:math/0505176.

Source Question

Morillon asks in the subsection "Questions", Question 3, PDF page 4, whether the Valdivia/Bourgain–Diestel/Bourgain–Fremlin–Talagrand weak* convex block compactness theorem remains true, in **ZF** or **ZFC**, when the hypothesis "no isomorphic copy of $\ell^1(\mathbb{N})$" is weakened to "no asymptotically isometric copy of $\ell^1(\mathbb{N})$".

Supporting Literature

Jimenez-Sevilla and Moreno, *A note on James' theorem for weak compactness*, arXiv:1508.00496, state in the final remarks, item (iii), PDF page 16:

> Pfitzner proved that if E does not contain asymptotically isometric copies of ℓ^1, then (B_{E^*}, ω^*) is convex block compact.

They cite H. Pfitzner, *Boundaries for Banach spaces determine weak compactness*, Invent. Math. 182 (2010), 585–604, page 601, Proposition 11.

Identification

Morillon's requested replacement is exactly:

$$E \text{ does not contain asymptotically isometric copies of } \ell^1(\mathbb{N})$$
$$\Longrightarrow$$
$$(B_{E^*}, w^*) \text{ is convex block compact.}$$

The Pfitzner proposition, as recorded in arXiv:1508.00496, proves precisely this implication in the usual ZFC setting. Therefore the ZFC version of Morillon's Question 3 has a positive answer.

Scope Limitations

This is not a new proof packet. It is a provenance packet identifying a later theorem that answers the extracted question after matching terminology.

The fully choiceless ZF version is not settled here. Morillon's Question 2, asking whether weak* block compactness can differ from weak* convex block compactness, also remains open here; arXiv:1508.00496 explicitly notes that this equivalence problem is still open in general.

Search Notes

The cheap run indexes had no prior packet for arXiv:math/0505176. Exact web searches on 2026-06-29 for the weak-block/convex-block and asymptotically-isometric ℓ^1 phrases did not surface a direct answer to Morillon's paper, but local source search found arXiv:1508.00496, whose final remarks identify Pfitzner's Proposition 11 as the exact ZFC strengthening.

References

- M. Morillon, *A new proof of James' sup theorem*, arXiv:math/0505176.
- M. Jimenez-Sevilla and J. P. Moreno, *A note on James' theorem for weak compactness*, arXiv:1508.00496.
- H. Pfitzner, *Boundaries for Banach spaces determine weak compactness*, Invent. Math. 182 (2010), 585–604.